

MyDesign Portfolio

Element I

	requirement	Priority level	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	The mechanism must clamp to the pole	High	1: Take open end of clamp and bring it toward the light pole 2: Fasten the clamp to pole	Fasten and unfasten clamp to pole (3 trials)	Clamp May: 1: stand strong 2: Fall downward	Clamp must hold onto pole
2	The mechanism must be sturdy.	High	1: Clamp the mechanism to the pole 2: attach board to the holder 3: apply pressure and see if the clamp is stable.	Fasten and then apply pressure (3 trials)	Clamp may: 1: bend in a different direction 2: Remain in the same position sturdy	Clamp must be sturdy enough to not bend or collapse
3	The mechanism must be adjustable	Medium	1: attach clamp to finger mold and pole 2: move clamp in different directions (left, right, up, down)	Push clamp one by one in all 4 different directions (3 trials)	Clamp may: 1: Move in the direction desired 2: Clamp may "resist" and not move	Clamp must be able to adjust in different directions
4	The mechanism must be cost effective	Low	1: Review the total costs 2: determine if it is under the budget	Review costs and see if it fits under budget (1 trial)	Clamp May: 1: cost more than 50\$ 2: cost less than 50\$	Clamp mechanism must cost under 50\$

We started off testing our 1st requirement high priority. The first requirement states that "The mechanism must clamp to the pole."

Requirement	Trial	Summary of trial	Time	Pass/Fail
Requirement 1	Trial 1	Matt placed the jaws of the clamp around the pole and tightened the clamp.	5.31seconds	Pass
Requirement 1	Trial 2	Matt placed the jaws of the clamp around the pole and tightened the clamp.	5.82 seconds	Pass
Requirement 1	Trial 3	Matt placed the jaws of the clamp around the pole and tightened the clamp.	4.97 seconds	Pass
Requirement 2	Trial 1	Matt applied pressure to the clamp after it was tightly fastened onto the pole.	3.18 seconds	Pass
Requirement 2	Trial 2	Matt applied pressure to the clamp after it was tightly fastened onto the pole.	2.50 seconds	Fail
Requirement 2	Trial 3	Matt applied pressure to the clamp after it was tightly fastened	3.21 seconds	Pass

		onto the pole.		
Requirement 3	Trial 1	Matt loosened the middle joint clamp piece and adjusted the board to various angles.	7.3 seconds	Pass
Requirement 3	Trial 2	Matt loosened the middle joint clamp piece and adjusted the board to various angles.	8.1 seconds	Pass
Requirement 3	Trial 3	Matt loosened the middle joint clamp piece and adjusted the board to various angles.	7.9 seconds	Pass
Requirement 4	Trial 1 (one trial only)	Matt checked the cost of our total combined element		

Based on our testing data, we found that the Square nub board holder (Final Design) had failed only one trial out of our whole testing plan; this means that the final iteration of the board holder was more sturdy because of the square nub.